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Differences in micromechanical properties of shales from different depositional environment: A case study of Longmaxi marine shale and Yanchang continental shale using nanoindentation

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ABSTRACT

An understanding of the mechanical properties of reservoir shale is of great significance for the efficient development of shale gas. Both marine and continental shale gas reservoirs in China have considerable development potential, but their different depositional environments may lead to substantial differences in their mechanical properties, which can result in production efficiency differences. In this study, nanoindentation, X-ray diffraction, backscattered electron imaging, and energy-dispersive X-ray spectroscopy were used to determine and analyze the mechanical properties and microtexture of marine and continental shale samples. The geogenesis of the microtexture of marine and continental shales and its influence on the mechanical properties were discussed. The results show that the elastic modulus of marine and continental shale samples are similar, but the hardness of latter is greater. The similar elastic modulus of the two shale samples may result from similar mineralogy. Due to differences in deposition and diagenesis, the marine shale sample forms a clay support matrix and the continental shale sample forms a rigid clastic support matrix, which results in lower hardness in the former and higher hardness in the latter. The low hardness of the shale with a clay support matrix indicates that it may be subject to more severe proppant embedment issue. The experimental results provide a useful reference for the development of these two types of shale gas reservoirs.

1. Introduction

Shale gas has attracted worldwide attention as a major alternative to conventional natural gas. (Chen et al., 2014; Liu et al., 2022; Lu et al., 2021; McGlade et al., 2013). The mechanical properties of the reservoir shale are important for shale gas exploration and development because of its impact on many engineering issues (e.g., hydraulic fracturing, proppant embedment, and wellbore stability) (Chen et al., 2022; Heng et al., 2020).

Due to the complex geological conditions of shale formation in China, both marine and continental organic-rich shale developed in the long-term tectonic evolution cycle (Wang et al., 2018; Zhang et al., 2016). The Longmaxi shale in the Sichuan Basin and Yanchang shale in the Ordos Basin are representatives of marine and continental shales in China, respectively, and are considered to have great potential for shale gas exploration and development (Dong et al., 2016; Tang et al., 2014; Zou et al., 2010). The petrophysical, geochemical, and geomechanical properties of marine and continental shales may differ due to the variable depositional environments (Li et al., 2018a). The differences between the petrophysical and geochemical properties of marine and continental shales and their implications for shale gas development have attracted the attention of researchers. Prior studies mainly focused on the differences in occurrence feature, mineralogy, geochemistry, pore structure, adsorption capacity, and gas storage capacity of marine and continental shales (Chen et al., 2016, 2019; Jiang et al., 2017; Li et al., 2018a; Wang and Guo, 2020; Xiao et al., 2021; Xie et al., 2021; Yang et al., 2017a, 2017b, 2021). The understanding of these differences is helpful to formulate targeted exploration and development plans.

However, the contribution to understanding the differences in mechanical properties between marine and continental reservoir shales is

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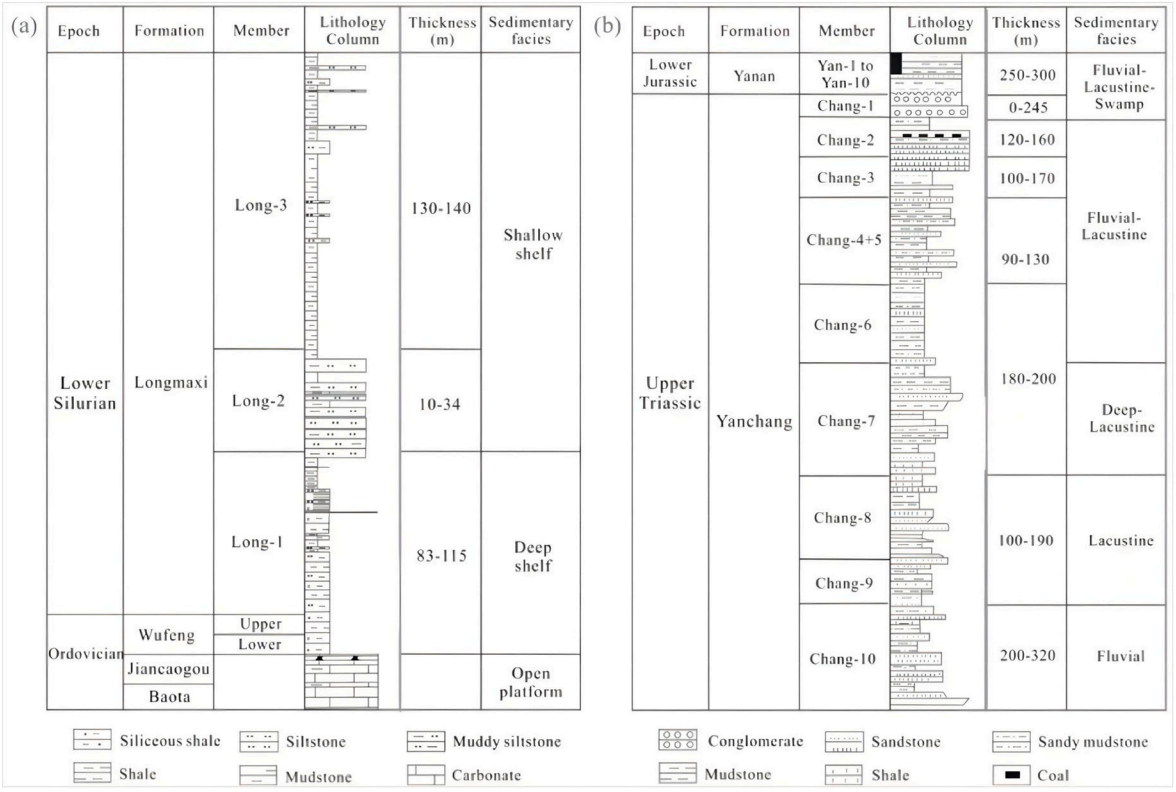


Fig. 1. (a) Stratigraphic column of the Lower Silurian Longmaxi Formation. (b) Stratigraphic column of the Upper Triassic Yanchang Formation (modified after (Li et al., 2018a)).

limited. Uniaxial and triaxial compression experiments are the most common way to obtain mechanical properties of rocks. Uniaxial or triaxial compression studies for shales were conducted mainly on outcrop shales, and only a few on reservoir shales. (Heng et al., 2020; Jia et al., 2021; Wang et al., 2020a). This is due to the scarcity of reservoir cores and the difficulty of making them into standard specimens. Application of nanoindentation technology in geological materials makes it easier to characterize the mechanical properties of reservoir shales (Ma et al., 2020). (Liu et al., 2016) and (Li et al., 2019) studied the mechanical properties of Bakken marine shale using nanoindentation and calculated its elastic modulus, hardness, and fracture toughness. Based on this method, Du et al. obtained meso-scale mechanical parameters such as elastic modulus, hardness, creep modulus, and fracture toughness of Longmaxi marine shale (Du et al., 2021). Wu et al. used nanoindentation to obtain mineral elastic moduli of Yanchang continental shale (Wu et al., 2020). However, these studies have been conducted independently around shales from only one depositional environment, without evaluating and analyzing the mechanical properties of shales from different depositional environments under the same experimental conditions. The role of depositional and diagenetic processes in controlling shale geomechanics property has been recognized. The work of Deng et al. shows that due to the difference of diagenetic process, the microtexture of shale in the upper and lower Longmaxi formation are different, which lead to the difference of geomechanical properties (Deng et al., 2021). Similar conclusions were reached by Delle Piane et al. in their study of the Goldwyer and Bongabini Formations shales in the Caning Basin, Australia (Delle Piane et al., 2015). There are great differences in sediment sources, sedimentation and diagenesis between marine and continental shale, which may lead to great geomechanical differences.

In this paper, we selected marine shale samples from the Lower Silurian Longmaxi formation in the Sichuan Basin and continental shale samples from the Upper Triassic Yanchang Formation in the Ordos Basin for testing and analysis. Nanoindentation, X-ray diffraction (XRD), backscattered electron imaging (BSE), and energy dispersive spectroscopy (EDS) were used to determine the micromechanical properties and microtexture of the samples. Furthermore, the influence of the depositional environment on the mechanical properties of the reservoir shale was discussed from a geological perspective. The impact of differences in the mechanical properties of marine and continental shales on shale gas development was also discussed.

2. Geological setting

Marine shale samples were taken from the Lower Silurian Longmaxi formation in Sichuan Basin, China. The Sichuan Basin is located in southwestern China, which is bounded by the fault zone of Longmen Mountain to the west, the fault zone of the Micang and Daba mountains to the north, and the Hunan-Guizhou-Hubei fold belt to the southeast (Wang et al., 2021). In the period of geological history, east of the Sichuan basin experienced the Caledonian movement, the Hercynian movement, the Indosinian movement, the Yanshan movement and the Himalayan movement, superposition and transformation of multiphase movements formed the present structural pattern (Li et al., 2018a). In Late Ordovician, Central Guizhou Uplift in the south, Sichuan Uplift in the west and Xuefeng Palaeo-high in the east turned the area from open sea in the Lower-Middle Ordovician into restricted sea surrounded by uplifts, bringing about the extensive lowenergy, undercompensated and anoxybiotic depositional environments. Two global transgressions happened at the end of Ordovician and at the beginning of Silurian and gave rise to the deposition of Longmaxi formation Organic-rich shales (Guo et al., 2014; Li et al., 2018a). Depositional indicators in the Longmaxi Formation suggest both deep and shallow water shelf settings (Fig. 1a). The lower member of the Longmaxi Formation consists mainly of thick, black, siliceous shale, calcareous-siliceous mixed shale and micritic shale with abundant fossils, including graptolites, sponge

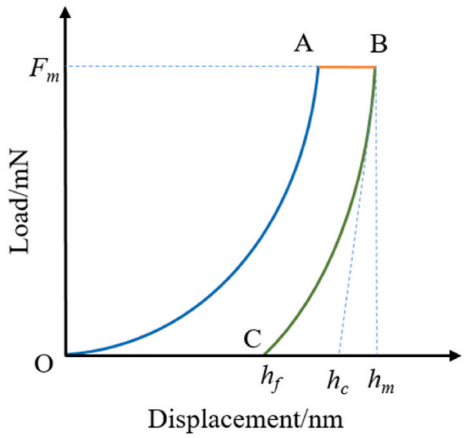


Fig. 2. Schematic diagram of a nanoindentation load-displacement curve.

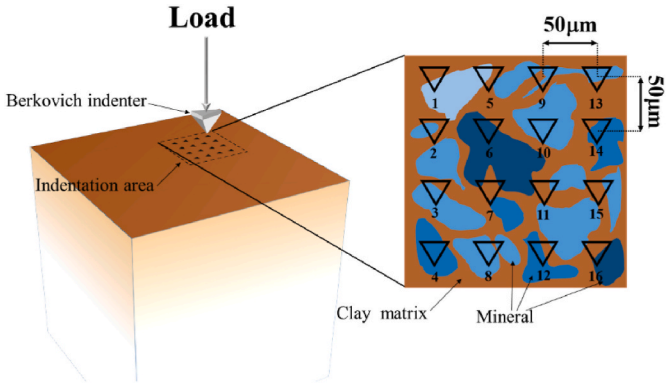


Fig. 3. Schematic diagram of the nanoindentation grid.

spicules, siliceous radiolarians and other planktonic algae. This assemblage indicates the deep water conditions and low sedimentation rates The upper member consists of argillaceous, laminated shale, argillaceous-calcareous mixed laminated shale and wavy bedded shale with a more depauperate fossil assemblage indicative of intermediate to shallow depositional conditions (Chen et al., 2016; Deng et al., 2021; Han et al., 2016). The Longmaxi shale has a low ratio of illite–smectite mixed layer versus illite, and is considered to be at a higher stage of diagenesis (Yang et al., 2017a).

Continental shale samples were taken from Upper Triassic Yanchang Formation (Chang-7 member) in Ordos Basin, China. The Ordos Basin is located at the tectonic joint of the eastern and western domains in China. It was a part of the Great North China Basin during the Paleozoic period. The Indosinian Movement during the Late Triassic caused the northern margin of the Yangtze Plate to collide with the North China plate, and the large-scale inland depression lake basin in Ordos was developed under the coupling effects of basin and mountain. The Basin is divided into six secondary tectonic units (termed as Western Margin Thrust Belt, Tianhuan Depression, Yishaan Slope, Jinxi Flexural Fold, Yimeng Uplift, and Weibei Uplift) according to the present structural morphology and evolution history of the basin (Fu et al., 2020). During the depositional period of the late Triassic Yanchang Formation, a large inland depression basin was formed under the influence of Indosinian movement. A set of terrigenous clastic rocks of Yanchang Formation, mainly composed of lacustrine and fluvial facies, was formed during the depositional period. The Yanchang Formation can be divided into 10 sections from top to bottom (Wang et al., 2020b; Zhang et al., 2021) (Fig. 1b). According to the lithology and deposition characteristics, the deposition period of Chang-7 member is the heyday of the lake, developing a wide range of deep and semi-deep lacustrine shales and mudstone with average thickness 70–80 m (Li et al., 2018a). The higher contents of illite–smectite mixed layer suggests that the Yanchang shale is in a low diagenetic stage (Yang et al., 2017a).

3. Experimental methods

3.1. Sample polishing

A rigorous polishing workflow is required to meet the sample surface flatness requirements for nanoindentation testing. The cores were cut into small cubes of 10 × 10 × 10 mm size by wire cutting. Mechanical polishing was then carried out on the surface perpendicular to the bedding. This workflow includes grinding with different grades of sandpaper (800 mesh, 1200 mesh, 2500 mesh, and 5000 mesh) and polishing with diamond suspension polishing fluid (1 μm and 0.5 μm) combined with flannelette. A Leica EM-RES 102 ion thinning instrument was then used for argon ion polishing. The shale samples were polished for 4 h at an accelerating voltage of 4 kV and polishing inclination of 3°.

3.2. XRD analyses

The rest of the sample was crushed into 60–80 mesh powder for XRD analysis to quantitatively analyze the mineral composition of shale samples. XRD analysis was performed on Bruker D8 ADVANCE following the Chinese Oil and Gas Industry Standard SY/T 5163–2018.

3.3. Nanoindentation

The principle of nanoindentation is that a diamond indenter with a specific shape and material parameters is vertically intruded into a sample surface and then withdrawn after reaching the set depth or load. The load and corresponding indentation depth are continuously recorded to obtain the load-displacement curve of the entire process, which can be used to determine mechanical parameters such as hardness and elastic modulus. A typical load-displacement curve is shown in Fig. 2. Section OA represents the loading stage, during which the deformation can be regarded as the superposition of elastic deformation and plastic deformation. Section AB represents the load holding stage and section BC section represents the unloading stage, the latter of which can be regarded as the elastic deformation recovery stage. The Oliver-Pharr method is a classical approach for calculating the hardness and elastic modulus of a sample (Oliver and Pharr, 1992):

$$E_r = \frac{\sqrt{\pi}S}{2\eta\sqrt{A_c}} \tag{1a}$$

$$H = \frac{F_{\max}}{A_c} \tag{1b}$$

where E_r is the reduced modulus, S is the contact stiffness, which is calculated by the initial stages of the unloading curve, $S = (dF/dh)_{h=h_{\max}}$. η is a constant related to the indenter geometry, for a Berkovich indenter, η = 1.034 (Pharr et al., 2011). H is the hardness, $F_{\max}$ is the peak load, and A_c is the contact area, which can be calculated as:

$$A_c = 24.5{h_c}^2 \tag{2}$$

$$h_c = h_{\max} - 0.75 \times \frac{F_{\max}}{S} \tag{3}$$

where h_c is the contact depth, $h_{\max}$ is the maximum depth. The reduced modulus can be converted into the elastic modulus according to:

$$\frac{1}{E_r} = \frac{1-\nu^2}{E} + \frac{1-\nu_{ind}^2}{E_{ind}} \tag{4}$$

where E and E_{ind} are elastic moduli of the sample and indenter, respectively, and ν and ν_{ind} are the Poisson's ratios of the sample and

Table 1
Mineralogy of two shale samples.

	quartz/ %	feldspar/ %	carbonate/ %	clay mineral/%	pyrite/ %
marine shale	40.3	4.7	10.8%	41.8	2.4
continental shale	34.4	11.9	11.9%	42.9	0

indenter, respectively. Previous studies have shown that a change of ν has little influence on the calculation results of E. We therefore set $\nu = 0.3$ for shale based on previous research experience. For a Berkovich indenter, $E_{ind} = 1141$ GPa and $\nu_{ind} = 0.07$(Liu et al., 2016).

Shale is a typical heterogeneous material that consists of a variety of mineral components, including quartz, feldspar, pyrite, and clay minerals. A single indentation therefore does not fully reflect the micro-mechanical properties of shale (Sun et al., 2020). In this study, nanoindentation was conducted over a 4 × 4 grid with 50-μm spacing. The grid was set in a pre-engraved box on the sample surface so that it could be quickly found for subsequent electron microscopy observations (Fig. 3). The scale of mechanical response is related to the applied load. The mechanical properties of the shale constituents can be obtained at low loads ($F_{max} \sim 5$ mN), while the bulk mechanical properties of the shale composite can be obtained at high loads (also called micro-indentation when $h_m > 1$ μm) (Charlton et al., 2021; Cheng et al., 2022; Kasyap and Senetakis, 2022; Katende et al., 2021b). In this study, a high load of 400 mN was applied with the aim of obtaining as homogenized mechanical response as possible. The quasi-static constant loading rate mode was applied with a loading rate of 10 mN/s. Upon reaching the maximum load, the maximum load was maintained for 15 s and then unloaded. A NanoTest Vantage system was used for the nanoindentation tests.

3.4. Micro-morphology observation

The micro-morphology was observed in the backscattering (BSE) mode of the scanning electron microscope. BSE is an imaging method of

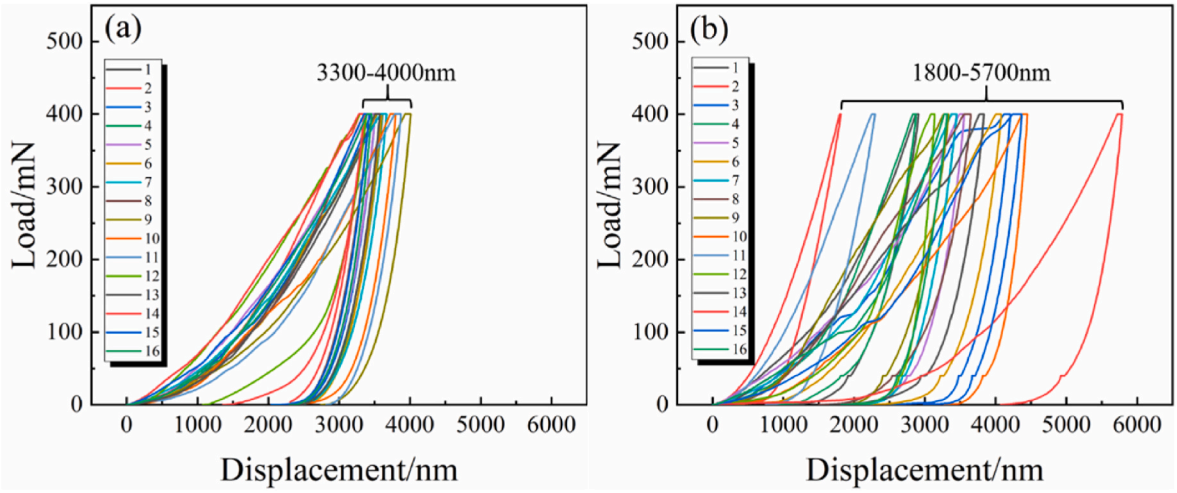


Fig. 4. Load-displacement curves of (a) marine shale sample and (b) continental shale sample.

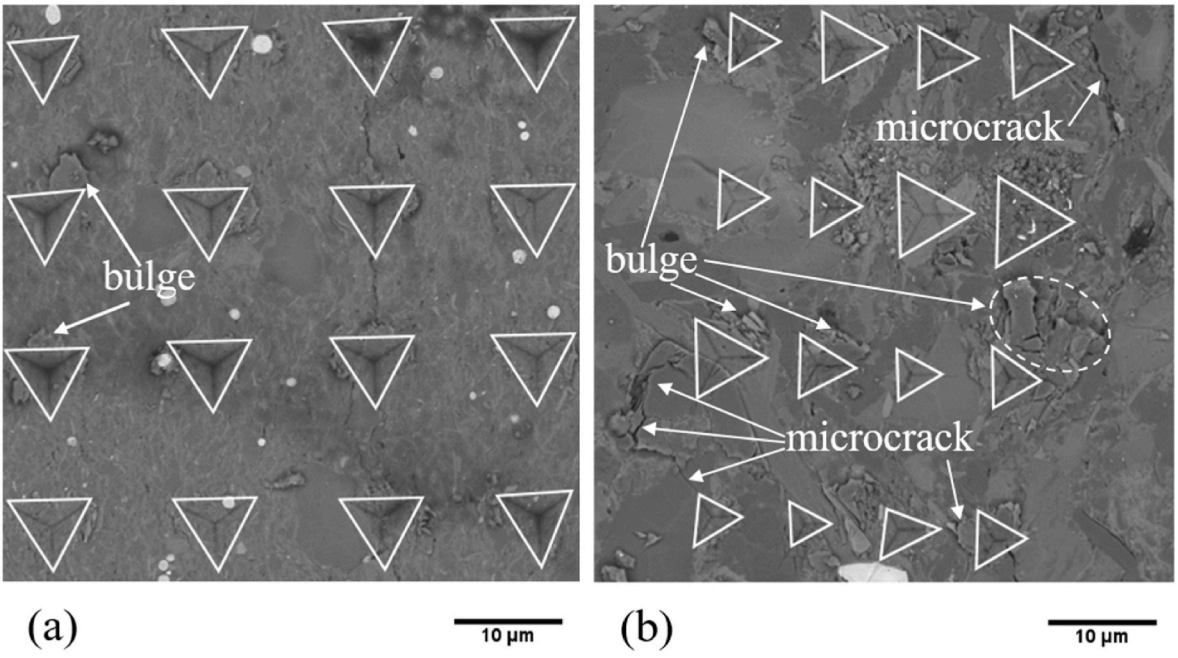


Fig. 5. Micro-morphology of (a) marine shale and (b) continental shale after indentation.

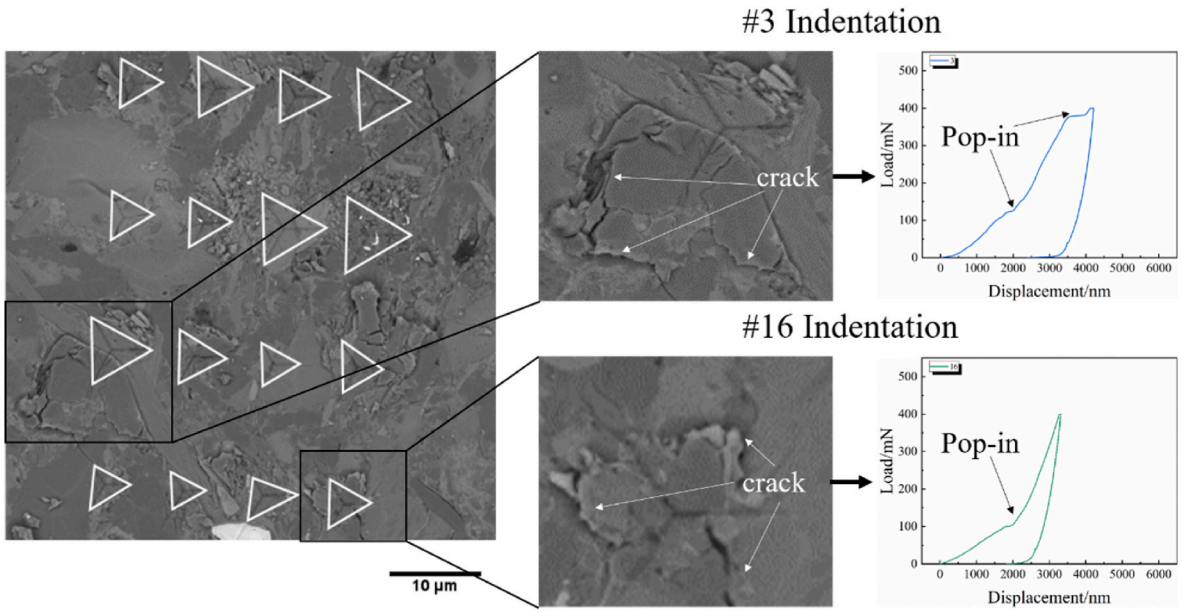


Fig. 6. Correspondence of electron microscope images and load-displacement curves. The load-displacement curves of the #3 and #16 indentations show notable pop-in phenomena, and considerable number of microcracks generated near the two indentations.

scanning electron microscope. Differing from secondary electron imaging, BSE imaging is based on the atomic number of the observed object. Minerals with different average relative molecular weights are reflected by relative brightness differences in a BSE image. BSE images can therefore be used to distinguish different mineral grains or aggregate crystals of different clay minerals. Energy dispersive spectroscopy (EDS) can be used to obtain information regarding the spatial distribution of different elements. The micro-morphology and mineral information of the indentation area can therefore be obtained using a combination of BSE imaging and EDS analysis (Houben et al., 2016). BSE and EDS were applied using a HITACHI TM4000PLUS II electron microscope and an Oxford energy dispersive spectrometer. The working distance is 5.8 mm and the acceleration voltage is 10 kV.

4. Results and discussion

4.1. Mineralogy

Although from completely different depositional environments, the mineralogy of the two shales are strikingly similar (Table 1). Both shales are clay-rich shales with over 40% clay content. The contents of felsic minerals in marine and continental shales are 45.0% and 46.3%, respectively, and the contents of carbonate minerals are 10.8% and 11.9%, respectively.

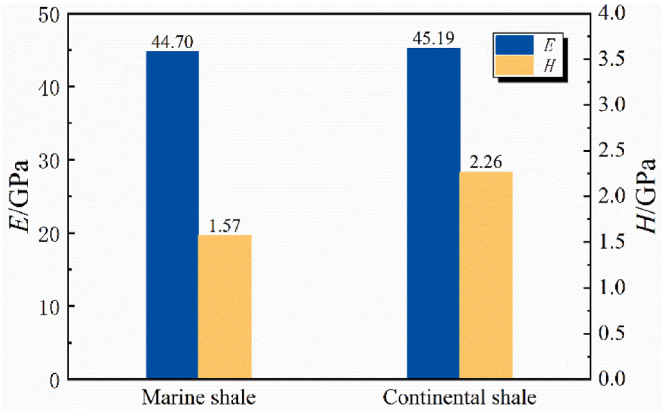


Fig. 7. Average elastic modulus and average hardness of marine shale and continental shale.

4.2. Load-displacement curve and indentation morphology

The load-displacement curves of the two types of shale are shown in Fig. 4. The serial number of the curves in Fig. 4 corresponds to the serial number of the indentations in Fig. 3. The load-displacement curves of marine shale are relatively concentrated, and the maximum displacement ranges between 3300 and 4000 nm and shows a good uniformity. In contrast, the load-displacement curves of continental shale are discrete and the maximum displacement ranges between 1800 and 5700 nm. This indicates that the mechanical response of marine shale shows good homogeneity under the loading conditions set in the experiment, whereas the load-displacement curves of continental shale show significant differences, which means that the mechanical responses differ between the different test points.

The indentation morphology of marine shale and continental shales are shown in Fig. 5. The indentation imprint of marine shale differs from that of continental shale. Firstly, the indentation imprints of marine shale are roughly equal in size, while those of continental shale differ. Equation (2) relates that the indentation imprint area is related to the indentation depth. Greater indentation depths are associated with larger residual indentation imprint areas, which usually implies weaker mechanical properties. There are also nearly no microcracks near the indentation imprint of the marine shale, whereas a large number of microcracks and bulges formed near the indentation imprint of the continental shale. In general, higher material brittleness makes it easier to form microcracks during indentation. Under the same load, microcracks were produced near the indentation imprint in the continental shale sample but not in the marine shale sample, which may indicate that the continental shale sample is more brittle.

A “pop-in” is the phenomenon of a displacement jump in the load-displacement curve. The local strain energy of a sample continuously increases during the loading process. The accumulated strain energy will eventually be released in one of three forms: (1) elastic deformation recovery; (2) plastic deformation dissipation; or (3) destruction of the internal sample structure (Manjunath and Jha, 2019). When the local strain energy of a sample accumulates to a critical value, grain

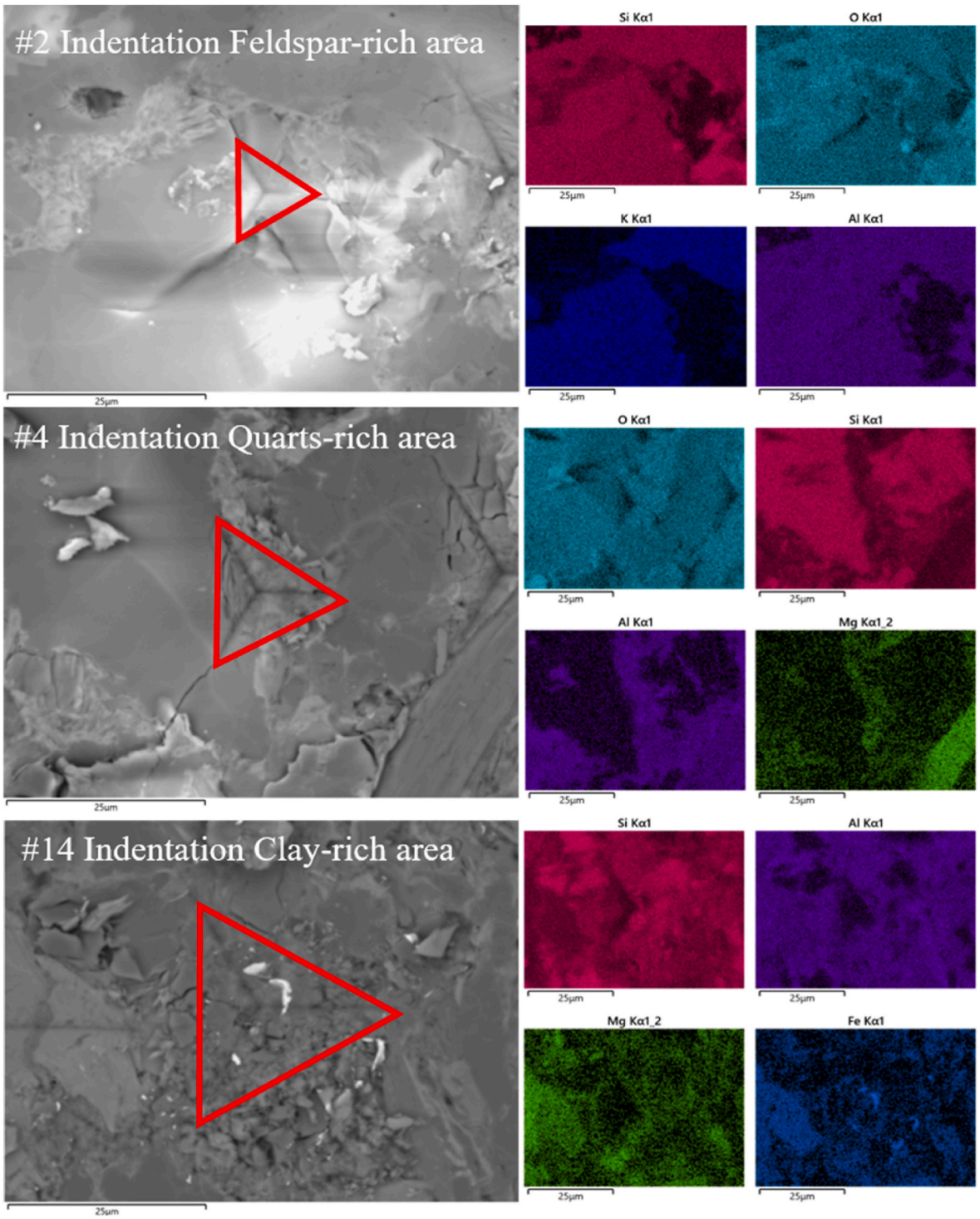


Fig. 8. EDS images of different indentation. The #2 indentation area is rich in Si, O, K, and Al and is identified as feldspar-rich area. The #4 indentation area is rich in Si and O and is identified as quartz-rich area. The #14 indentation area is rich in Si, Al, Mg, and Fe and is identified as a clay-rich area.

dislocation, pore collapse, or crack propagation may occur, resulting in a sudden displacement jump. More brittle material is generally more likely to exhibit the pop-in phenomenon. The loading section of the load-displacement curves of marine shale increases smoothly without a pop-in, whereas pop-ins appear in some of the continental shale curves (Fig. 6). The load-displacement curves of the #3 and #16 indentations of the continental shale and the morphology of their corresponding positions. Considerable number of microcracks generated near the #3 and #16 indentations, which can explain the pop-ins in their corresponding curves. The indentation morphology indicates that the marine shale sample has a more ductile mechanical response, while the continental shale sample has a more brittle mechanical response.

4.3. Hardness and elastic modulus

The average elastic modulus and average hardness of the two shale samples are shown in Fig. 7. The elastic modulus of the marine shale samples ranges from 34.97 to 48.91 GPa with an average of 44.70 GPa,

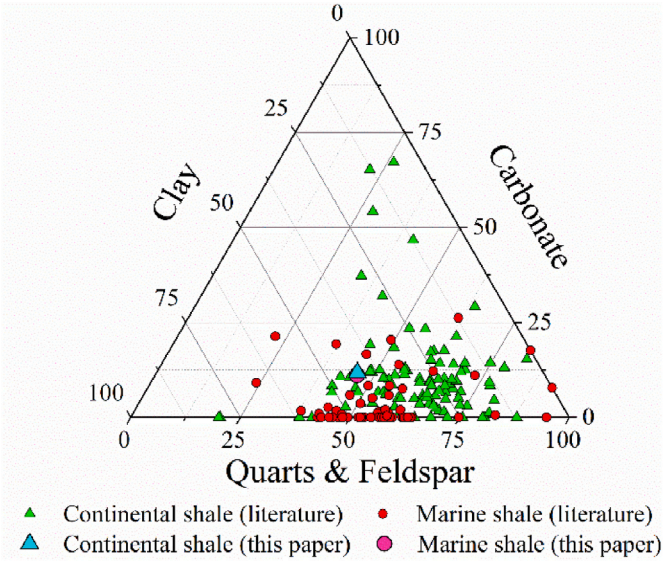


Fig. 9. The ternary diagram for marine and continental shales (data from Jiang et al., 2016; Li et al., 2018b; Xiao et al., 2021; Yang et al., 2017a; Yang et al., 2017b; Yang et al., 2018).

and the hardness ranges from 1.19 to 1.91 GPa with an average of 1.57 GPa. The elastic modulus of the continental shale samples ranges from 25.55 to 80.27 GPa with an average of 45.19 GPa, and the hardness ranges from 0.92 to 9.25 GPa with an average of 2.26 GPa. Detailed mechanical result is provided in Appendix A. The results show that the marine shale sample and continental shale sample have similar average elastic moduli, but the average hardness value of the continental shale sample is significantly higher than that of the marine shale sample.

4.4. Influential factors of the elastic modulus

The factors affecting the elastic modulus of shale are complex, including mineral composition and microtexture. There are significant differences in the elastic modulus of different constituent minerals. In this paper, EDS was used to correspond the mechanical parameters to the mineralogy of shale. The samples after nanoindentation were analyzed by EDS to obtain the surface distribution of elements in the samples. The mineral information under each indentation spot was then obtained according to the chemical formula of the common shale minerals. For example, areas dominated by silicon and oxygen enrichment are considered quartz, and areas with potassium, silicon, aluminum, and oxygen enrichment are considered K-feldspar (Houben et al., 2016; Shi et al., 2020). In the continental shale samples #2, #4, and #14 indentations were identified as feldspar-rich, quartz-rich, and clay-rich area, respectively (Fig. 8), corresponding to an elastic modulus of 76.04 GPa, 45.72 GPa, and 25.54 GPa, respectively. The bulk elastic property of the shale composite is the comprehensive embodiment of the elastic properties of these constituent minerals.

The highly similar elastic modulus and mineralogy of the marine and continental shale samples in this study may indicate that, at least at the microscopic scale, mineralogy controls the elastic modulus of the shale. This is supported by the results of prior mineralogical and micromechanical property investigations. The study of Du et al. showed that the mineralogy and elastic modulus of the clay-rich Longmaxi shale are similar to the results presented here (Du et al., 2021).

The similarity of elastic modulus between marine and continental shale samples in this paper cannot be simply generalized to all marine and continental shale reservoirs because of the limited number of samples in this paper. The mineralogy of a broader range of marine and continental shales were investigated through literature research (Fig. 9). These samples came from a wide range of marine and continental depositional basins, including Sichuan Basin, upper Yangtze Basin, Ordos Basin, Bohai Bay Basin, etc (Jiang et al., 2016; Li et al., 2018b; Xiao et al., 2021; Yang et al., 2017a, 2017b, 2018). The wide range of mineralogical variability in both marine and continental shales may indicate that their modulus of elasticity is also highly variable.

Prior study showed that in addition to mineralogy, microtexture differences due to depositional and diagenetic processes could also affect the elastic modulus (Deng et al., 2021). However, in this study, microtexture differences appear to have a greater effect on shale hardness than on elastic modulus, which is discussed in detail in the next section.

4.5. Influential factors of hardness

Although the similarity in mineralogy between marine and continental shale samples in this study is a fortuitous result, it provides a research case of the effect of microtexture on the mechanical properties of shales excluding the influence of mineralogy. Fig. 10 shows the distinct microtexture of marine and continental shale samples. For the marine shale sample, clay minerals have strong preferential orientation, and fine-grained quartz and feldspar mainly exist in the clay matrix as discrete grains. For the continental shale sample, rigid grains (e.g., quartz and feldspar) have large grain sizes and contact each other to form a rigid support matrix. The clay minerals fill the intergranular spaces of the rigid particles in an amorphous form.

The microtexture of shale is the result of original depositional environment and subsequent diagenesis. The upper part of the Longmaxi Formation is interpreted to represent a relatively shallow water anoxic

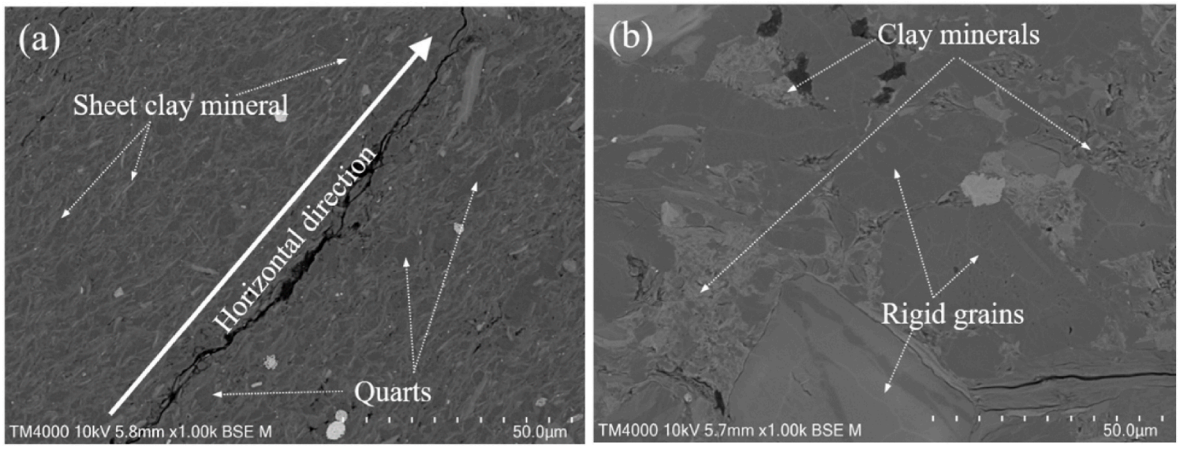


Fig. 10. Microtexture of the (a) marine shale and (b) continental shale samples.

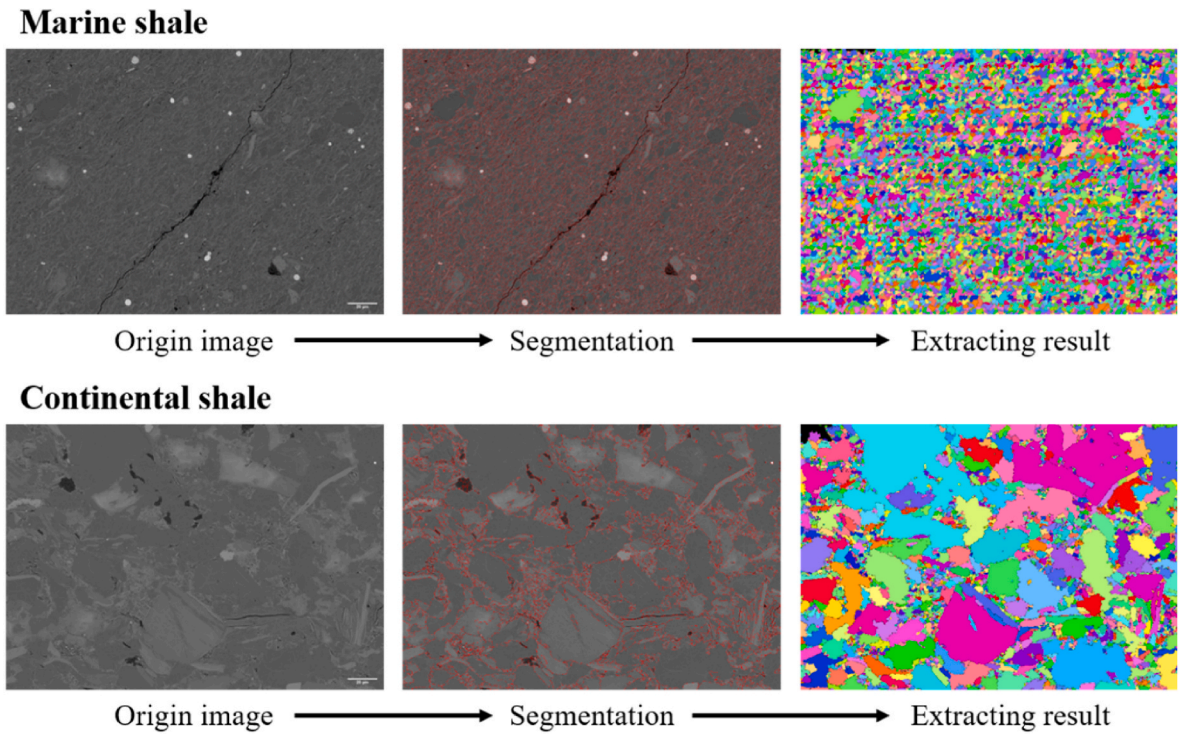


Fig. 11. Schematic diagram of mineral segmentation process.

shelf environment with minor fluctuations in depositional conditions. The primary composition of shale samples from this part contains a prominent fraction of clay minerals and a significant component of silt-size extrabasinal debris (Wang et al., 2014). In the following mechanical compaction process during burial, platy clay particles in the primary composition of shale samples were re-oriented and re-arranged under overburden with the fabric of preferred orientation of clay minerals parallel to the overburden, where silt-sized extrabasinal debris is surrounded by a matrix containing abundant alignment clay-size clay minerals (Deng et al., 2021; Wang et al., 2021). During the transformation of smectite into mixed-layer illite-smectite, silica will be released and precipitated into authigenic fine-grained quartz within the clay matrix as discrete grains (Nadeau et al., 2002). The above processes culminate in a microtexture of marine shale sample with a clay matrix and fine-grained discrete grains. During the depositional period of Yanchang formation, the Ordos Basin entered the inland basin evolution stage. A large number of coarse-grained terrigenous clasts were carried into the basin from the provenance area around the basin by water, and a set of fluvial lacustrine facies and delta clastic deposits with a thickness of more than 1000 m were deposited (He et al., 2019; Ruppel et al., 2017). The terrigenous clasts acted as a rigid skeleton during burial, resisting the influence of mechanical compaction and later diagenetic changes, forming a rigid support matrix.

The particle size of the constituent minerals is also an influential factor in the mechanical response of the shale (Sabri et al., 2016). The mineral particle size of continental shale is usually larger than that of marine shale (Xiao et al., 2021).

To further quantify the particle size difference between the two types of shale, an advanced method using ImageJ software was adopted to obtain particle size information from the BSE images. Different minerals are known to exhibit different brightness in BSE images with observable boundaries. We therefore applied the watershed algorithm to segment the minerals. Within the set threshold range, the watershed algorithm can recognize the closed area with the same brightness as a mineral particle (Chandra and Vishal, 2021; Sarkar et al., 2018). After the mineral segmentation, the statistics of the particle size information can be executed. All of the involved algorithm codes are integrated into a plug-in called MorphoLibJ, and a visual GUI is provided. In this study, 750 μm × 540 μm areas were selected from the marine shale and continental shale images to analyze the particle size information. The segmentation process is shown in Fig. 11.

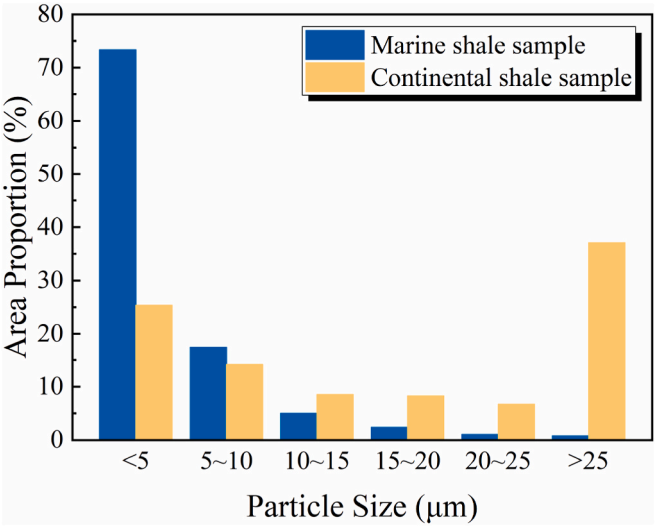


Fig. 12. Proportion of minerals with different particle sizes.

The equivalent radius was used to characterize the particle size, which is defined as:

$$R_E = \sqrt{\frac{A}{\pi}} \tag{5}$$

where R_E and A are the equivalent radius and area of a mineral particle, respectively.

The proportion of mineral particles with different equivalent radii is shown in Fig. 12. The marine shale sample mainly contains mineral particles with equivalent radii of <5 μm. In contrast, the dominant

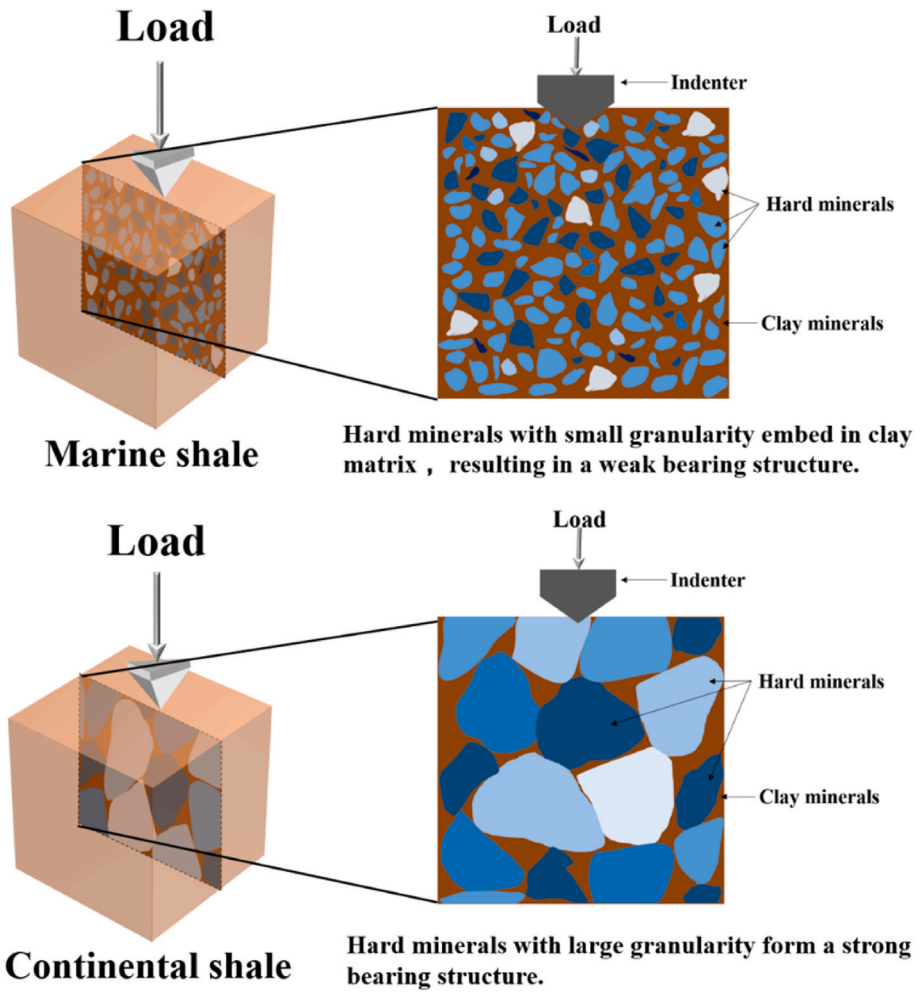


Fig. 13. A simplified conceptual model to explain the effect of microtexture and particle size on shale hardness.

equivalent radius of the continental shale minerals is larger. Mineral particles larger than 25 μm in size predominate。

In terms of the effect on hardness, the function of hard minerals is similar to that of coarse aggregate in concrete, which acts as a bearing framework. The particle size of the hard minerals in marine shale is small and embedded within a clay matrix. The dispersed hard mineral particles cannot form an effective load-bearing structure. When the indenter intrudes, the clay matrix is prone to plastic deformation, which poorly restricts the movement or rotation of the hard embedded minerals. Hard minerals are prone to dislocate with the dislocation of clay minerals, resulting in the formation of a weak bearing structure of the marine shale and poor resistance to the indenter penetration (i.e. low hardness). In contrast, the hard minerals of the continental shale are larger in size and closely arranged, which restricts their movement or rotation during indenter penetration and forms a rigid bearing structure. The continental shale therefore has a high resistance to indenter penetration (i.e., high hardness). A simplified conceptual model is proposed to explain the effect of microtexture and particle size on shale hardness (Fig. 13).

The influence of deposition and diagenesis on shale microstructure, and furthermore on geomechanics, has been confirmed by other macromechanical studies. Delle piane et al. investigated the geomechanical properties of shales from the Bongabinni and Goldwyer formations in the Caning Basin, Western Australia, using triaxial tests and ultrasonic velocity tests, and analyzed the effects of diagenesis. The Goldwyer Formation shales form a clay support matrix, while the dolomite and anhydrite cementation in the Bongabinni Formation shales provide a rigid support structure. Therefore, the Bongabinni Formation is mechanically stronger and stiffer than the Goldwyer Formation (Delle Piane et al., 2015). The work of Deng et al. shows that due to the difference of diagenesis, the upper member of Longmaxi formation forms clay support matrix and the lower member of Longmaxi formation forms quartz support matrix. In the triaxial compression experiment, the former shows elastic-plastic mechanical response, while the latter shows elastic mechanical response (Deng et al., 2021). In general, rigid particle supported matrix is mechanically stronger and stiffer than clay supported matrix. This paper extends this understanding to micro scale.

4.6. *Implications for shale gas development*

The hardness difference between marine and continental reservoirs must be considered in shale gas production to estimate the possible problems and put forward solutions. Hardness is a parameter that characterizes a material's ability to resist external invasion, which is consistent with the ability of reservoir rocks to resist proppant embedment (Katende et al., 2021a, 2021b). Lower hardness leads to more severe proppant embedment. The experimental results therefore show that proppant embedment in marine shale reservoirs with clay support matrix may be more severe than that in continental reservoirs. This adverse effect will be further amplified in a water-based fracturing fluid system (Khan et al., 2021; Lyu et al., 2020). The proppant and injection scheme must therefore be contrapuntally selected in the fracturing design.

5. Conclusion

The micromechanical properties of reservoir shales from the Longmaxi Formation in the Sichuan Basin and the Yanchang Formation in the Ordos Basin were studied using nanoindentation. The geogenesis of the microtexture of marine and continental shales and its influence on the mechanical properties were discussed.

The results show that the elastic modulus of Longmaxi marine shale is similar to that of Yanchang continental shale, but the hardness of the latter is greater. The similar elastic modulus of the two shale samples may result from similar mineralogy as a coincidence. Due to differences in deposition and diagenesis, the marine shale samples form a clay support matrix and the continental shale samples form a rigid clastic supported matrix, which results in lower hardness in the former and higher hardness in the latter. The low hardness of the shale with a clay support matrix indicates that it may be subject to more severe proppant embedment issue.

Credit author statement

Yiyu Lu: Conceptualization, Funding acquisition, **Qi Cheng**: Investigation, Writing – original draft, **Jiren Tang**: Conceptualization, Supervision, **Wenchuan Liu**: Writing – review & editing, **Honglian Li**: Writing – review & editing, **Jie Liu**: Investigation, **Zijie Xu**: Investigation, **RongRong Tian**: Investigation, **Xiao Sun**: Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix

Table 1
Mechanical result of marine and continental shale.

	marine shale		continental shale	
	H (GPa)	*E* (GPa)	*H* (GPa)	*E* (GPa)
1	1.51	44.2	2.71	42.3
2	1.78	48.4	9.25	76.0
3	1.60	47.4	1.08	37.3
4	1.51	45.3	2.72	45.7
5	1.61	46.0	1.57	43.2
6	1.51	46.7	1.18	34.7
7	1.46	42.4	1.71	42.1
8	1.71	46.0	1.60	32.6
9	1.19	41.1	1.92	38.9
10	1.34	44.1	0.92	46.7
11	1.29	41.8	4.93	55.8
12	1.91	35.0	1.89	80.3
13	1.54	45.6	1.35	37.3
14	1.85	41.0	0.55	25.5
15	1.69	48.9	0.99	37.1
16	1.63	51.2	1.84	47.4
mean	1.57	44.7	2.26	45.2
SD	0.20	3.9	2.13	14.6

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